Minimax: three moves ahead

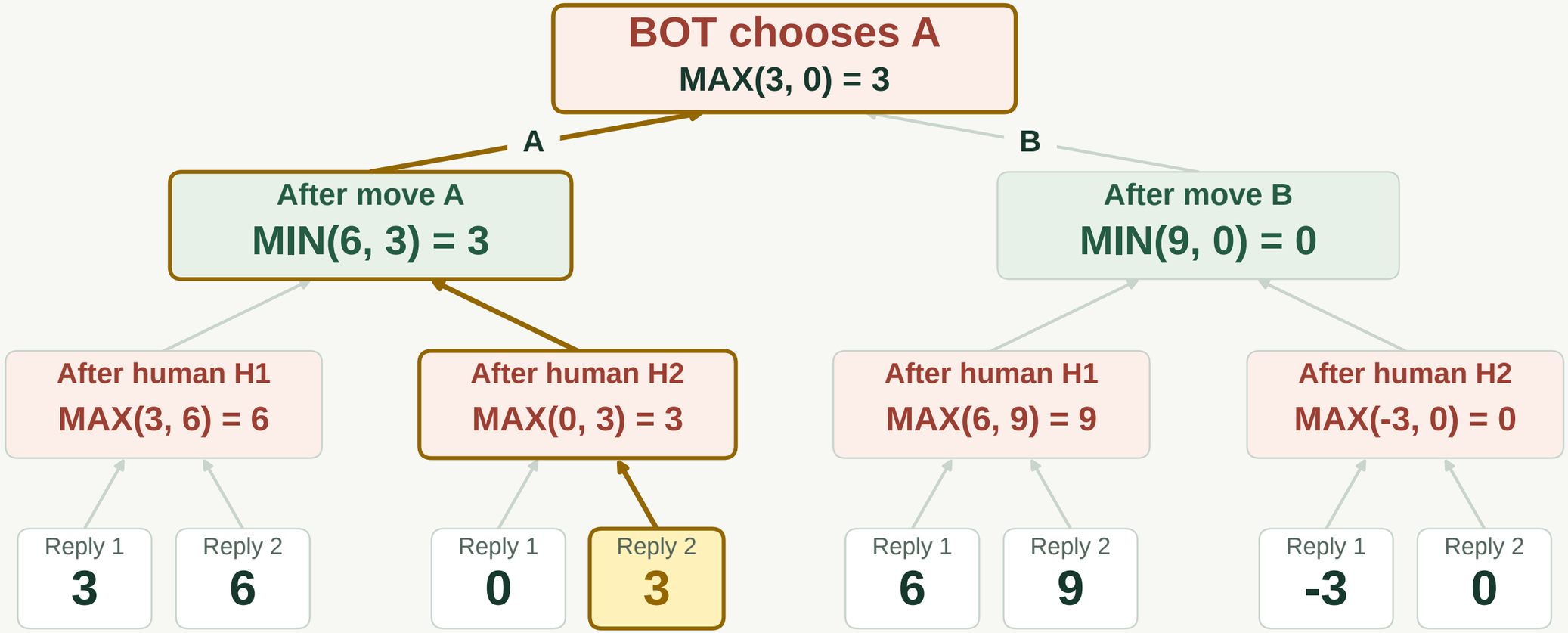
Explore BOT → HUMAN → BOT. Then return the scores upward to choose the first move.

1 BOT
MAX
findBestMove()
depth = 3

2 HUMAN
MIN
minimax()
depth = 2

3 BOT
MAX
minimax()
depth = 1

SCORE
EVALUATE
evaluateBoard()
depth = 0



A returns 3. B returns 0. findBestMove() returns A.

Gold traces the predicted choices: A → H2 → Reply 2. Only the first move A is played now.

Illustrative scores; alternating turns and no early winner. Every score is from the BOT's perspective.

The same example, one path per row

The score is calculated after the third simulated move. Then MAX and MIN return values upward.

1 BOT first move	2 HUMAN response	3 BOT final response	Board score · depth 0
A	H1	Reply 1	**3**
A	H1	Reply 2	**6**
A	H2	Reply 1	**0**
A	**H2**	**Reply 2**	**3**
B	H1	Reply 1	**6**
B	H1	Reply 2	**9**
B	H2	Reply 1	**-3**
B	H2	Reply 2	**0**

A = MIN(MAX(3, 6), MAX(0, 3))
= MIN(6, 3) = 3

B = MIN(MAX(6, 9), MAX(-3, 0))
= MIN(9, 0) = 0

findBestMove(): MAX(3, 0) = 3 → choose A

Illustrative scores; alternating turns and no early winner. Every score is from the BOT’s perspective.